Native replay of the completed degree-83 panel

2026-10-12. Correct exact model: `icv1-f2m83-t6151469093347-9eba6cd5`. This descriptive report records independent arithmetic replay of completed outputs, not new measurements of algorithm performance.

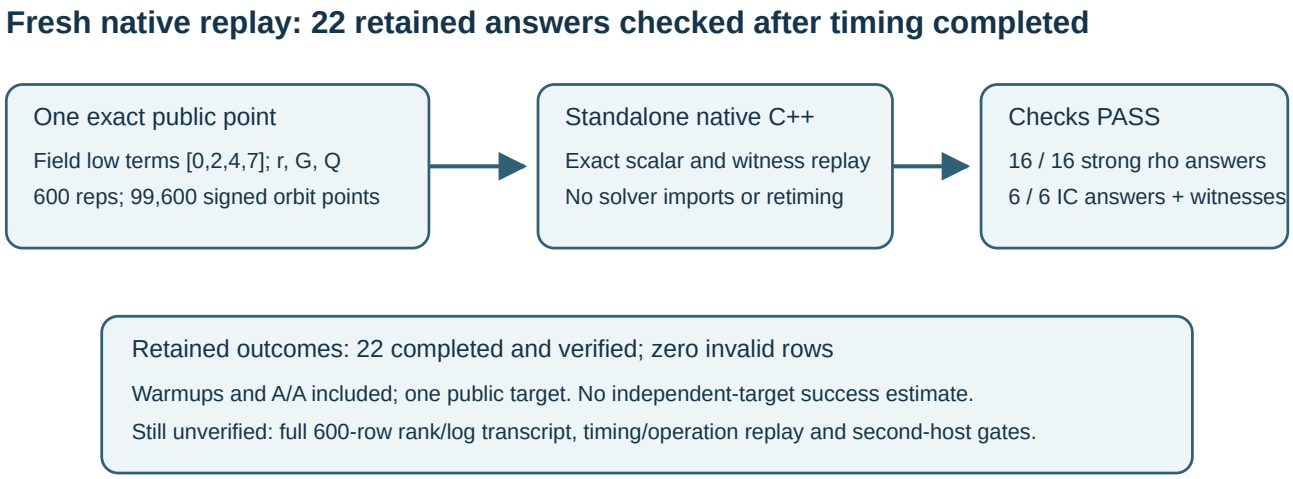


Figure 1: The native verifier checked the common instance and every recorded answer after all timed children completed. Counts include warmups and A/A controls and do not estimate success on independent targets.

Observed verification result

The parent evaluation task ran `check_fresh` after all timed children finished. The native receipt reports `PASS`, **22 independently verified answers**, zero invalid rows and 22 retained process dispositions with status `COMPLETED_VERIFIED`:

Producer arm	Retained rows	Scalar replay to exact public Q	Four-point witnesses
Strong signed-Frobenius rho	16	16/16	Not applicable
IC	6	6/6	6/6 re-add to Q

The counts include one warmup for each arm, ten strong A/A runs, and five formal matched repeats for each arm. All rows concern the same public point and are preserved individually. These are archive-replay counts, not an independent-target completion estimate or a speedup claim. Failed outcomes, caps and missing answers would remain explicit dispositions; none occurred in the checked 22 rows. The parent reports checker execution of approximately 0.62 seconds, outside all producer timing intervals. The checker was not rerun during this source/receipt review.

Exact arithmetic scope

The retained field is defined by `z^83+z^7+z^4+z^2+1`; the curve is $y^2+xy=x^3+x^2+1$. The native checker revalidates field irreducibility, prime subgroup `r=8569786107849059`, curve order and cofactor, generator/target membership, Frobenius eigenvalue and the complete retained factor base. The validated base has 600 representatives and 99,600 distinct signed-orbit points with correct scalar coefficient labels.

Every fresh recovered scalar passes a standalone polynomial-field affine replay `[d]G=Q` against these exact coordinates:

```
G_x = 66142446513981848602920l5
G_y = 88682604100292493915395l9
Q_x = 26558983455371661805329l0
Q_y = 38815263079916732652926l5
```

For every IC row the checker also validates four point indices against the independently checked base, compares x codes and intermediate abscissae, and re-adds the actual signed points to Q. The validation-only fixture scalar is compared only after a recovered answer passes scalar replay. This verifier did not execute either research solver and was not a solver input. It uses an independent C++ implementation of the standard field and curve arithmetic, importing no research solver. The fresh check ran on the same host after the fresh panel, not on an independent performance host.

Identity correction and evidence boundaries

The correct full explicit-field model identity is:

`ICV1:f2m-83-a5fd0d39:6151469093347:`
`967140655691088192855606 2:0x1:-7:unk:r:9eba6cd5d359`

The line break above is for rendering only. The older manually inserted alias `icv1-f2m83-t6151469093347-cdcc5432` names a different modulus `[0,1,2,45]`. The original archive receipt and archive report remain unchanged and must be read with `CURVE_ID_ERRATUM.md`. Their explicit arithmetic inputs were correct; the identity-label correction does not change their point checks. The fresh checker checks the explicit modulus, subgroup, G and Q and depends on neither alias.

The 600 coefficient coordinates are formal factor-log bookkeeping, not 600 independent group generators. The rational subgroup is cyclic, generated by G. Each `R_j=[L_j]G`; the guided inhomogeneous system recovers 600 scalar values L_j relative to G's known logarithm one. `INVARIANTS.md` explains the coefficient map, orbit phase transport, root-index witness and strong rho's two-coefficient collision equation with implementation references.

Correct point answers do not independently certify the complete rank construction. The retained target and summary rows do not expose all 600 rank equations, their decomposition witnesses, the final matrix and solved factor-log vector. Independent replay of that preparation transcript remains unverified. The native replay also does not reproduce operation counts, validate host-isolation promotion gates or certify timings on another host.

Both fresh producers retain positive timer residuals, separate from canonicalization's runs of zero bits. In formal pair 1, rho's full interval exceeds its phase sum by 401 ns; IC's full target interval exceeds its five phase sum by 0.029956 ms. Full online intervals include those residuals. No exact phase partition is asserted by this verification report.

Durable evidence and reproduction

- Fresh native receipt: `fresh_replay_receipt.json`.
- Independent native sources: `check_fresh.cpp` and included `check.cpp`.
- Original archive verification: `receipt.json`, `REPORT.md`, `REPORT.pdf`.
- Identity correction: `curve_id_erratum.json`, `CURVE_ID_ERRATUM.md`.
- Existing algorithm explanation: `INVARIANTS.md`.
- Input/source/executable provenance: the three preserved SHA-256 files.

Fresh receipt SHA-256:

`980a3d7d52db96a559dd36ad12e317c9b12be9f1c161a2d0d86803aafcc72441`

Compile/replay instructions are in `README.md`. The editable verification diagram is `fresh_verification.svg`, with its vector PDF rendering. The parent evaluation task owns the main study report, timed records and statistical analysis. This copy adds source/receipt verification documentation without altering those artifacts or claiming repository publication.